

Hands-on Assignments for Reporting Aggregated Data Using the Group Functions

Complete the below hands-on assignments before proceeding with the next Topic

Update Completion Status

No.	Cat	Hands-on Assignment	Topics Covered
1	M	Find the highest, lowest, sum, and average salary of all employees. Label the columns	Group Functions
2	M	Maximum, Minimum, Sum, and Average, respectively. Round your results to the nearest whole number.	Group Functions

Oracle SQL Developer : Oracle

Find Database Object

Oracle

- ☒ All Schemas
- ☒ All Object Types
- ☐ All Identifier Types
- ☐ All Identifier Usages
- ☐ Columns
- ☐ Code
- ☐ All Dependencies

Go

Worksheet

```
SELECT ROUND(MAX(SALARY)) AS "Maximum",  
       ROUND(MIN(SALARY)) AS "Minimum",  
       ROUND(SUM(SALARY)) AS "Sum",  
       ROUND(AVG(SALARY)) AS "Average"  
FROM HR.EMPLOYEES;
```

Script Output

Task completed in 0.21 seconds

Maximum	Minimum	Sum	Average
24000	2100	691416	6462

3 M

Modify the above query to display the minimum, maximum, sum, and average salary for each job type.

Group by Clause

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Worksheet

Query Builder

```

SELECT JOB_ID,
       ROUND(MIN(SALARY)) AS "Minimum",
       ROUND(MAX(SALARY)) AS "Maximum",
       ROUND(SUM(SALARY)) AS "Sum",
       ROUND(AVG(SALARY)) AS "Average"
FROM HR.EMPLOYEES
GROUP BY JOB_ID
ORDER BY JOB_ID;
  
```

Script Output

Task completed in 0.181 seconds

JOB_ID	Minimum	Maximum	Sum	Average
AC_ACCOUNT	8300	8300	8300	8300
AC_MGR	12008	12008	12008	12008
AD_ASST	4400	4400	4400	4400
AD_PRES	24000	24000	24000	24000
AD_VP	17000	17000	34000	17000
FI_ACCOUNT	6900	9000	39600	7920
FI_MGR	12008	12008	12008	12008
HR_REP	6500	6500	6500	6500
IT_PROG	4200	9000	28800	5760
MK_MAN	13000	13000	13000	13000
MK_REP	6000	6000	6000	6000
JOB_ID	Minimum	Maximum	Sum	Average
PR_REP	10000	10000	10000	10000
PU_CLERK	2500	3100	13900	2780
PU_MAN	11000	11000	11000	11000
SA_MAN	10500	14000	61000	12200
SA_REP	6100	11500	250500	8350
SH_CLERK	2500	4200	64300	3215
ST_CLERK	2100	3600	55700	2785
ST_MAN	5800	8200	36400	7280

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet

hr_install.sql

0.142 seconds

```
SELECT JOB_ID,
COUNT(*) AS "Number of People"
FROM HR.EMPLOYEES
GROUP BY JOB_ID
ORDER BY JOB_ID;
```

Script Output

Task completed in 0.142 seconds

JOB_ID	Number of People
AC_ACCOUNT	1
AC_MGR	1
AD_ASST	1
AD PRES	1
AD_VP	2
FI_ACCOUNT	5
FI_MGR	1
HR_REP	1
IT_PROG	5
MK_MAN	1
MK_REP	1
PR_REP	1
PU_CLERK	5
PU_MAN	1
SA_MAN	5
SA_REP	30
SH_CLERK	20
ST_CLERK	20
ST_MAN	5

19 rows selected.

5

M

Determine the number of managers without listing them. Label the column Number of Managers

Group by Clause

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet

hr_install.sql

0.13600001 seconds

```
SELECT COUNT(DISTINCT MANAGER_ID) AS "Number of Managers"
FROM HR.EMPLOYEES
WHERE MANAGER_ID IS NOT NULL;
```

Script Output

Task completed in 0.136 seconds

Number of Managers
18

6

M

Find the difference between the highest and lowest salaries. Label the column DIFFERENCE.

Group Functions

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Worksheet

Query Builder

```
SELECT MAX(SALARY) - MIN(SALARY) AS "DIFFERENCE"
FROM HR.EMPLOYEES;
```

Script Output

Task completed in 0.127 seconds

DIFFERENCE
21900

7

M

Create a report to display the manager number and the salary of the lowest-paid employee for that manager. Exclude anyone whose manager is not known. Exclude any groups where the minimum salary is \$2000 or less. Sort the output in descending order of salary.

Where Clause,
Group by Clause,
Having Clause,
Order by Clause

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Worksheet

Query Builder

```
SELECT MANAGER_ID AS "Manager Number",
       MIN(SALARY) AS "Lowest Salary"
FROM HR.EMPLOYEES
WHERE MANAGER_ID IS NOT NULL
GROUP BY MANAGER_ID
HAVING MIN(SALARY) > 2000
ORDER BY MIN(SALARY) DESC;
```

Script Output

Task completed in 0.198 seconds

Manager Number	Lowest Salary
102	9000
205	8300
146	7000
145	7000
108	6900
149	6200
147	6200
148	6100
201	6000
100	5800
101	4400
103	4200
124	2500
123	2500
114	2500
120	2200
122	2200
121	2100

18 rows selected.

8

M

Create a query to display the total number of employees and, of that total, the number of employees hired in 1980, 1981, and 1982. Create appropriate column headings.

Single Row and
Group Functions

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet

Query Builder

```
SELECT COUNT(*) AS "Total Employees",  
       SUM(CASE WHEN EXTRACT(YEAR FROM HIRE_DATE) = 1980 THEN 1 ELSE 0 END) AS "Hired 1980",  
       SUM(CASE WHEN EXTRACT(YEAR FROM HIRE_DATE) = 1981 THEN 1 ELSE 0 END) AS "Hired 1981",  
       SUM(CASE WHEN EXTRACT(YEAR FROM HIRE_DATE) = 1982 THEN 1 ELSE 0 END) AS "Hired 1982"  
FROM HR.EMPLOYEES;
```

Script Output

Task completed in 0.139 seconds

Total Employees	Hired 1980	Hired 1981	Hired 1982
107	0	0	0